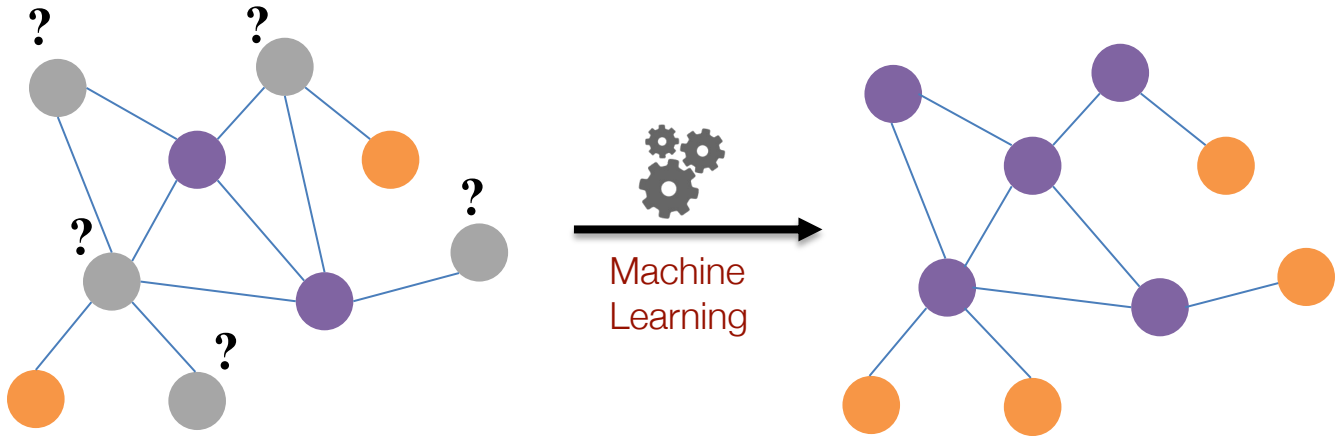


Machine Learning with Networks

Classical ML tasks in networks:

- Node classification
 - Predict a type of a given node
- Link prediction
 - Predict whether two nodes are linked
- Community detection
 - Identify densely linked clusters of nodes
- Network similarity
 - How similar are two (sub)networks

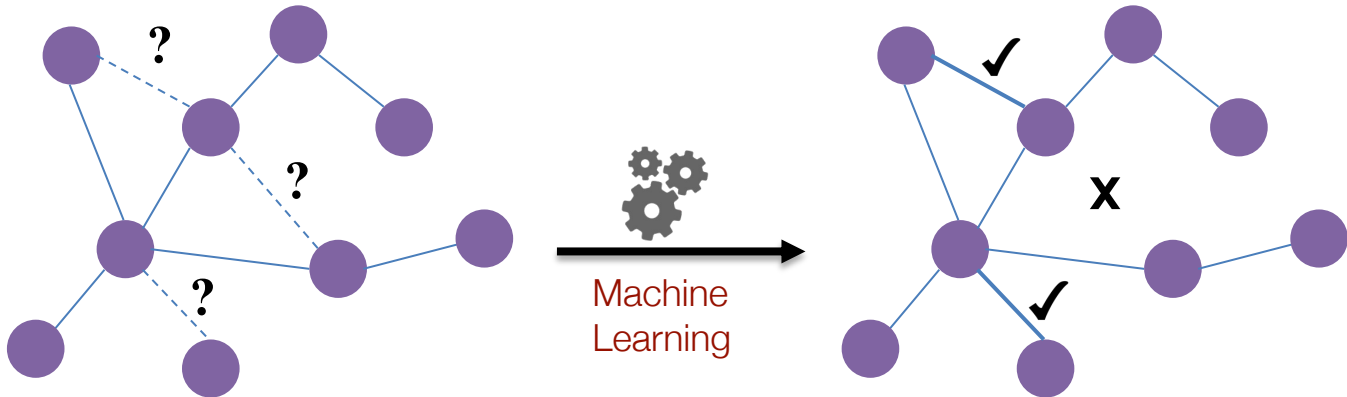
Example: Node Classification



Classifying the function of proteins in the interactome!:

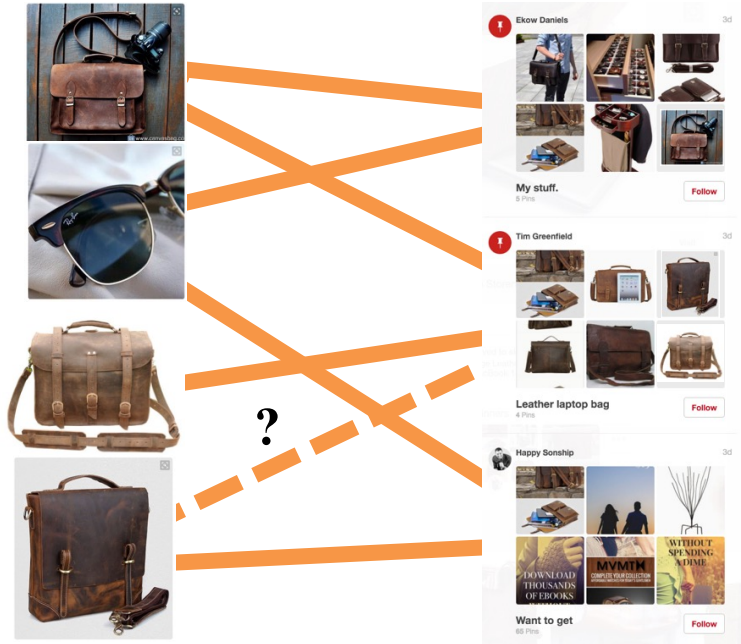


Example: Link Prediction



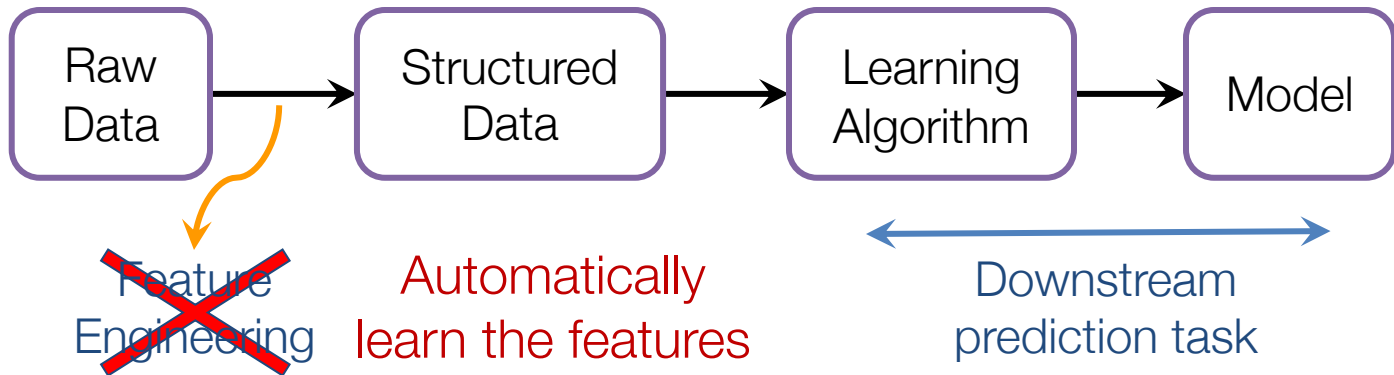
Example: Link Prediction

Content
recommendation is
link prediction!



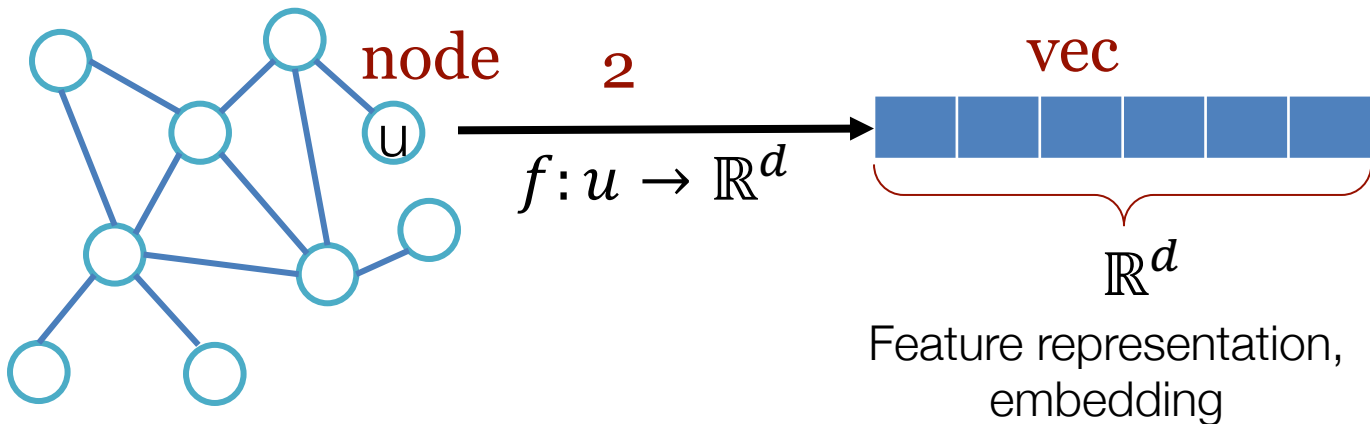
Machine Learning Lifecycle

- (Supervised) Machine Learning Lifecycle: This feature, that feature.
Every single time!



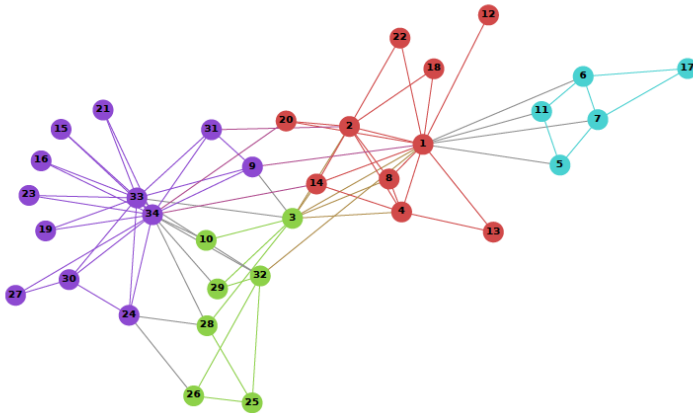
Feature Learning in Graphs

Goal: Efficient task-independent feature learning for machine learning in networks!

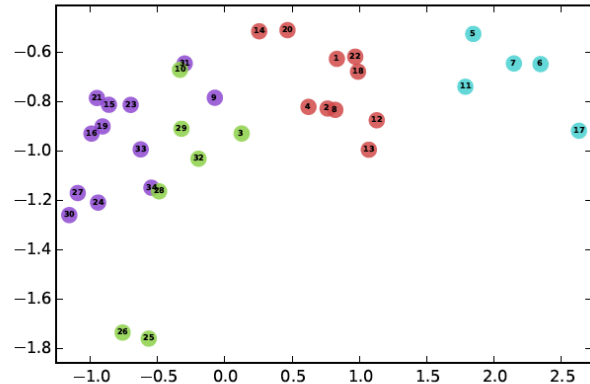


Example

- Zachary's Karate Club Network:



Input

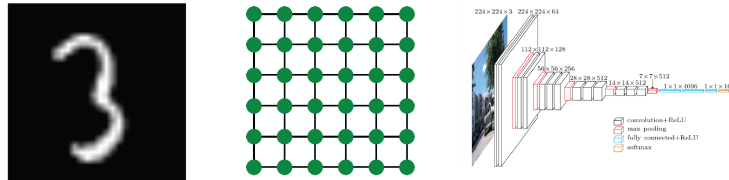


Output

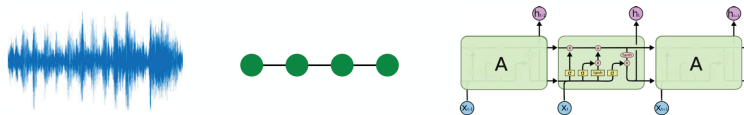
Image from: [Perozzi et al. 2014](#). DeepWalk: Online Learning of Social Representations. *KDD*.

Why Is It Hard?

- Modern deep learning toolbox is designed for simple sequences or grids.
 - CNNs for fixed-size images/grids....



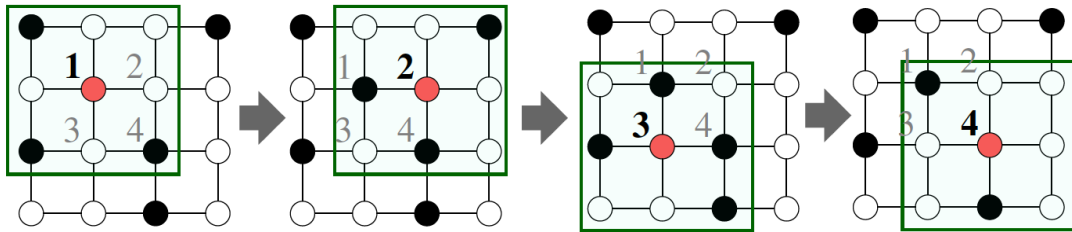
- RNNs or word2vec for text/sequences...



Why Is It Hard?

- But networks are far more complex!

- Complex topographical structure
(i.e., no spatial locality like grids)



- No fixed node ordering or reference point
(i.e., the isomorphism problem)
- Often dynamic and have multimodal features.

Outline

- 1) Node embeddings
 - Map nodes to low-dimensional embeddings.
- 2) Graph neural networks
 - Deep learning architectures for graph-structured data
- 3) Applications